Lingrui Lin, Ph.D. student



I am a Ph.D. student from the School of Astronomy and Space Science of Nanjing University (NJU). I am very easy going and open minded, so please feel free to contact me.
[*Last update: October 6, 2026*]

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Education

Sep 2021 – present	**Ph.D. student, School of Astronomy and Space Science, Nanjing University** Thesis: *Multi-scale gas dynamics in galaxies* Expected completion: June 2027 Supervisor: *Zhi-Yu Zhang (NJU)* Research hosts and collaborators: *Federico Lelli (INAF), Carlos De Breuck (ESO)*
Sep 2017 – Jun 2021	**B.Sc., School of Astronomy and Space Science, Nanjing University** Thesis: *Kinematics of atomic gas in supernova remnant IC 443* Supervisor: *Zhi-Yu Zhang*

Research Experience

Sep 8, 2025 – Sep 9, 2026	**Visiting Ph.D. student (funded by CSC),** Arcetri Astrophysical Observatory (INAF), Florence, Italy Topic: *Cold gas dynamics of rotating disk galaxies at* $z = 4$*–5* TRICEPS (Tracing Resolved Ionized Carbon Emission in Primeval Systems): second author of the survey paper (I); first author of the kinematic paper (III), both under review at *Astronomy & Astrophysics* Host: *Federico Lelli*
July 26, 2025 – Aug 2, 2025	**1st China Astrochemistry Summer School,** Xiamen University, Xiamen, China Poster: *Measuring H I/H_2 in Galactic Low-Metallicity Molecular Clouds*; poster prize
July 7, 2025 – July 11, 2025	**JWST Data and Science Summer School,** Westlake University, Hangzhou, China Selected participant and speaker; hands-on training in JWST data reduction and processing
Oct 1, 2023 – Dec 22, 2023	**Early-Career Scientific Visitor,** European Southern Observatory (ESO), Garching bei München, Germany Topic: *[C I] kinematics of a starburst radio galaxy at cosmic noon* Host: *Carlos De Breuck*
Sep 23, 2023 – Sep 30, 2023	**Visitor,** Arcetri Astrophysical Observatory (INAF), Florence, Italy Topic: *[C I] kinematics of a starburst radio galaxy at cosmic noon* Host: *Federico Lelli*

Publications

Refereed papers: first-authored

[1] **L. Lin**, Z.-Y. Zhang, J. Wang, *et al.*, "Inadequate turbulent support in low-metallicity molecular clouds," *Nature Astronomy*, vol. 9, pp. 406–416, Mar. 2025. DOI: 10.1038/s41550-024-02440-3.

[2] **L. Lin**, F. Lelli, C. De Breuck, *et al.*, "Gas dynamics in an AGN-host galaxy at $z \simeq 2.6$: Regular rotation, noncircular motions, and mass models," *Astron. Astrophys.*, vol. 693, A91, A91, Jan. 2025. DOI: 10.1051/0004-6361/202450814.

Refereed papers: coauthored

[3] G. Luo, Z.-Y. Zhang, T. G. Bisbas, *et al.*, "Anomalously high deuterium fractionation in a galactic translucent cloud: A challenge to chemical models," *Astron. Astrophys.*, vol. 712, L17, p. L17, Aug. 2026. DOI: 10.1051/0004-6361/202661031.

[4] Z. Lu, J. Wang, **L. Lin**, *et al.*, "Sharp line profile of radio recombination line H29α in W49A," *Publ. Astron. Soc. Aus.*, vol. 43, e030, e030, Feb. 2026. DOI: 10.1017/pasa.2026.10165.

[5] XSMT Project Collaboration Group, Y. Ao, J. Chang, *et al.*, "Scientific Objectives of the Xue-shan-mu-chang 15-meter Submillimeter Telescope," *Science China Physics, Mechanics & Astronomy*, Sep. 2025, Accepted; arXiv:2509.13983. DOI: 10.48550/arXiv.2509.13983.

[6] W. Wang, C. De Breuck, D. Wylezalek, *et al.*, "JWST + ALMA ubiquitously discover companion systems within $\lesssim$18 kpc around four $z \approx 3.5$ luminous radio-loud AGN," *Astron. Astrophys.*, vol. 696, A88, A88, Apr. 2025. DOI: 10.1051/0004-6361/202553668.

[7] H.-T. Huang, A. W. S. Man, F. Lelli, *et al.*, "Molecular Gas Mass Measurements of an Active, Starburst Galaxy at $z \approx 2.6$ Using ALMA Observations of the [C I], CO, and Dust Emission," *Astrophys. J.*, vol. 977, no. 2, 251, p. 251, Dec. 2024. DOI: 10.3847/1538-4357/ad9021.

[8] G. Luo, D. Li, Z.-Y. Zhang, *et al.*, "The CO-dark molecular gas in the cold H I arc," *Astron. Astrophys.*, vol. 685, L12, p. L12, May 2024. DOI: 10.1051/0004-6361/202450067.

[9] Y. Sun, Z.-Y. Zhang, J. Wang, *et al.*, "An improved method to measure ^{12}C/^{13}C and ^{14}N/^{15}N abundance ratios: revisiting CN isotopologues in the Galactic outer disc," *Mon. Not. R. Astron. Soc.*, vol. 527, no. 3, pp. 8151–8192, Jan. 2024. DOI: 10.1093/mnras/stad3643.

[10] F. Li, Z.-Y. Zhang, J. Wang, G. Luo, **L. Lin**, and J. Zhou, "Dense gas properties around the centre of the Circinus galaxy," *Mon. Not. R. Astron. Soc.*, vol. 527, no. 1, pp. 531–543, Jan. 2024. DOI: 10.1093/mnras/stad3241.

[11] L. Zhang, Z.-Y. Zhang, J. W. Nightingale, *et al.*, "Discovery of a radio jet in the Cloverleaf quasar at z = 2.56," *Mon. Not. R. Astron. Soc.*, vol. 524, no. 3, pp. 3671–3682, Sep. 2023. DOI: 10.1093/mnras/stad2069.

[12] F. Lelli, Z.-Y. Zhang, T. G. Bisbas, *et al.*, "Cold gas disks in main-sequence galaxies at cosmic noon: Low turbulence, flat rotation curves, and disk-halo degeneracy," *Astron. Astrophys.*, vol. 672, A106, A106, Apr. 2023. DOI: 10.1051/0004-6361/202245105.

[13] Y. Deng, Z.-Y. Zhang, P. Zhou, *et al.*, "Multiple gas phases in supernova remnant IC 443: mapping shocked H_2 with VLT/KMOS," *Mon. Not. R. Astron. Soc.*, vol. 518, no. 2, pp. 2320–2340, Jan. 2023. DOI: 10.1093/mnras/stac3139.

[14] G. Luo, Z.-Y. Zhang, T. G. Bisbas, *et al.*, "Dependence of Chemical Abundance on the Cosmic-Ray Ionization Rate in IC 348," *Astrophys. J.*, vol. 942, no. 2, 101, p. 101, Jan. 2023. DOI: 10.3847/1538-4357/aca657.

Under review

[15] **L. Lin**, F. Lelli, C. Bacchini, *et al.*, "TRICEPS. III. Cold gas disks in massive galaxies at $z = 4$–5: fast rotation and low pressure support," Astronomy & Astrophysics, under review, 2026.

[16] F. Lelli, **L. Lin**, Z.-Y. Zhang, *et al.*, "TRICEPS. I. Tracing cold gas, dust, and stars at sub-kpc scales in massive star-forming galaxies at $z = 4$–5," Astronomy & Astrophysics, under review, Sep. 2026. arXiv: 2609.30375 [astro-ph.GA].

Others

[17] **L. Lin**, Z.-Y. Zhang, and J. Wang, *The initial condition for star formation varies with metallicity: insights from the Galactic outer disk molecular clouds*, Chinese Science Bulletin, 70, 3367–3369, Scientific commentary, in Chinese, 2025. DOI: 10.1360/CSB-2025-0454.

[18] C. De Breuck, B. Emonts, W. Wang, *et al.*, "ALMA observations of high redshift radio galaxies," in *ALMA at 10 years: Past, Present, and Future*, Dec. 2023, 50, p. 50. DOI: 10.5281/zenodo.10244516.

Research Coverage

Feb 2025	INAF / Media INAF, Italy (Italian press coverage of Lin et al. 2025, *Nature Astronomy*)
Jan 2025	Aristotle University of Thessaloniki, Greece (Greek press release)
	Nature Astronomy (English Research Briefing)
	Nanjing University, China (Chinese press release)

Meetings & Conferences

July 10, 2026	Contributed talk, "Dynamically cold gas disks in massive galaxies at $z = 4$–5 from the TRICEPS survey", "How stars move around: what do they tell us about galaxies?", Sexten, Italy
July 3, 2026	ePoster with flash talk, "Dynamics of massive galaxy disks at $z = 4$–5 from the TRICEPS survey", EAS 2026 (SS3a; poster 1817), Lausanne, Switzerland
July 1, 2026	Contributed talk, "Gas and dust in massive galaxies at $z = 4$–5 from the TRICEPS survey", EAS 2026, Lausanne, Switzerland
June 16, 2026	Contributed talk, "Cold gas disks in massive galaxies at $z = 4$–5: fast rotation and low pressure support", Zoom-in Views of Galaxy Disks Across Cosmic Time, Nordita, Stockholm, Sweden
May 21, 2026	Galaxy Evolution Coffee, ESO
Jan 14, 2026	EG meeting, INAF-Arcetri
Nov 14, 2024	**Invited Talk,** Annual conference of molecular cloud and star formation, GXU
Dec 2023	Galaxy Evolution Coffee, ESO

Observational Projects

As Principal Investigator

ALMA	*An ACA census of Galactic metal-poor molecular clouds* 2021.2.00175.S; ACA total power, Band 6; 64.1 hours
NOEMA	*Resolving the gravity-dominated molecular clouds in the outer Galaxy* W23AY; PolyFiX; 8 hours
SMA	*Galactic metal-poor molecular clouds on sub-pc scales* 2022A-S042; 230/345-GHz receivers, Dual Rx correlator mode; two tracks
IRAM 30-m	*Pebble-sized dust grains in Galactic metal-poor molecular clouds* 105-20; NIKA2; 8.2 hours
JCMT	*Exploring the low dust emissivity index of Galactic metal-poor molecular clouds* M21BP041; SCUBA-2; 0.5 hours
SMT	*Dynamic states of molecular clouds in the outer Galaxy* Lin_L_22B_1; 1.3-mm receiver; 48 hours
FAST	*Measuring H I Fraction in Galactic Metal-poor Molecular Clouds* PT2021_0055; 19-beam receiver/Spec(W+N); 17 hours
	A HINSA census towards metal-poor molecular clouds in the Galactic far outer disk PT2024_0068; 19-beam receiver/Spec(W+N); 17.5 hours
GBT 100-m	*Resolving the turbulence-deficient molecular clouds in the outer Galaxy* GBT24B-167; Argus/VEGAS; 21 hours
Effelsberg	*CO-dark molecular gas in the outer Galaxy* 98-22; P180mm/XFFTS; 46.2 hours

As an active collaborator (selected)

ALMA	*Cold gas and dark matter in massive galaxies at $z = 4$–5* CoI, 2023.1.01724.S and 2025.1.00827.S; PI Federico Lelli
	Ultradeep [C II] observations of rotating gas disks at $z = 4.5$ CoI, 2023.1.00835.S, 2024.1.00839.S, 2025.1.01398.S, and 2026.1.00563.S; PI Federico Lelli
VLA	*Cold gas and dark matter in massive galaxies at $z = 4$–5* CoI, 24A-296 and 25A-307; WIDAR; 101.5 and 109.9 hours respectively; PI Federico Lelli

Observational Projects (continued)

JWST	*Luminous and dark matter in massive galaxies at $z = 4$–5* Cycle 2, program 3954; PI Federico Lelli
IRAM 30-m	*Dynamic states of molecular clouds in the outer Galaxy* Lead proposal contributor, 102-22 (2022); PI Thomas Bisbas
JCMT	*Dust in shocked regions of Supernova remnant IC 443* CoI and lead proposal contributor, M22AF002 (2022)
	Sub-virial molecular clouds in the outer Milky Way on core to parsec scales: Short-spacing for the SMA CoI and lead proposal contributor, M22BP027; HARP and $\bar{\text{U}}\bar{\text{u}}$ receivers/ACSIS; 25 hours; PI Hauyu Baobab Liu
FAST	*Atomic Clouds Driven by the Galactic Nuclear Wind* PT2024_0083; PI Jing Zhou
GBT 100-m	*An OH and CH survey of the molecular clouds in the outer-disk of the Milky Way* GBT23A-246; PI Gan Luo
	Mapping the CO-dark molecular gas in the outer Galaxy (2023) Lead proposal contributor

Awards & Grants

2026 – present	Zhujian Program for Doctoral Students, Nanjing University
Sep 8, 2025 – Sep 9, 2026	China Scholarship Council support for the research visit to INAF–Arcetri
2025	National Scholarship (China)
	Poster prize, 1st China Astrochemistry Summer School, Xiamen University
2021	Linqiao Scholarship (Linbridge Fund, Douglas Nelson Chao Lin)
	Outstanding Undergraduate Graduate
	President's Special Scholarship for Doctoral Students
2018 – 2020	Top Talent Program Scholarship $\times$ 3
2019	Annual Scholarship of National Astronomical Observatory, Chinese Academy of Sciences
	Renmin Scholarship
2018	National Encouragement Scholarship
	National College Mathematics Competition (Second prize, Non-Mathematics Major)

Responsibilities

Fall 2026	Organizer, "Discussions on the frontiers of high-z galaxies and the ISM", Nanjing University
Spring 2025	Teaching Assistant of "Astronomical Literature — Reading and Writing", NJU Astronomy
Spring 2024	Teaching Assistant of "Interstellar Medium", NJU Astronomy
2022 – 2023	Organization Member, Graduate Student Committee of NJU Astronomy
2018 – 2021	Monitor, NJU Astronomy Class 2017
2018 – 2019	Leader, Academic Division of NJU Great Astronomy Fans Association (NJU GAFA)
	Captain, NJU Astronomy Table Tennis team
2017 – 2018	Publicity Department Member, Student Union of NJU Astronomy
	Representative for Studies, NJU Astronomy Class 2017

Skills

Astronomical Expertise

Radio astronomy	Observing preparation; remote observations with SMT, GBT, and Effelsberg
	Spectral-line and continuum data reduction for ALMA, VLA, NOEMA, SMA, IRAM 30-m, JCMT, SMT, GBT, FAST, and Effelsberg
	Calibration, continuum subtraction, and spectral-line and continuum imaging using CASA
Gas dynamics	Three-dimensional disk fitting, rotation-curve mass modeling, and analysis of spectral cubes and velocity fields
Radio astronomy	Single dish and interferometry data combination in U-V plane
Astronomical Tools	*Proficient usage:* GILDAS/CLASS, GILDAS/Mapping, CASA, CARTA, DS9, ^{3D}BAROLO, `vcdisk`, Astropy, Photutils, MONTAGE
	Basic usage: GILDAS/PIIC, Starlink, MIR, Glue
Code development	**See more on GitHub: Astronlin**
	Highlight: (a) A Python FAST H I calibration pipeline; *(b)* MCMC-based HINSA fitting tools; *(c)* interactive scientific visualizations

Technical Skills

Coding	Python (proficient), Fortran (basic), C++ (basic), IDL (basic)
Operating System	Linux (I help maintain the Ubuntu/CentOS servers and remote-access services in the NJU group), macOS, Windows
Writing and editing	Vim (proficient), LaTeX (proficient), Microsoft Word (proficient)

Miscellaneous

Languages	Mandarin Chinese (Native), English (Fluent)
Sports	Table tennis (ping-pong; advanced), Swimming (proficient), Basketball (competent), Badminton (competent), Tennis (novice)
Photograph	Adobe Photoshop/Camera Raw, Adobe Premiere